# *CivicPulse*

## *FYP Code: 23F-90*

## *Software Requirement Specification*

BY

AYOUSH KUMAR, CMS. ID: 023-23-0220

SARICHAND, CMS. ID: 023-23-0024

RAINUKA BAI, CMS. ID: 023-23-0053

**Supervised by: Dr. Saif Hassan** _____

**Co-Supervised by: Dr. Ismail** _____

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF SCIENCE AND INFORMATION TECHNOLOGY

SUKKUR IBA UNIVERSITY

NOVEMBER 2026

# Document Revision History

| Version | Date | Prepared by | Description |
|---|---|---|---|
| 1.0 | November 2026 | Ayoush Kumar, Sarichand, Rainuka Bai | Initial version for supervisor review |

# Table of Contents

1. INTRODUCTION ..... 1
1.1. Purpose of Document ..... 1
1.2. Intended Audience ..... 1
1.3. Document Convention, abbreviations, and acronyms ..... 1
2. OVERALL SYSTEM DESCRIPTION ..... 2
2.1. Project Background ..... 2
2.2. Project Scope ..... 2
2.3. Not in Scope ..... 2
2.4. Project Objectives ..... 3
2.5. Stakeholders ..... 3
2.6. Operating Environment ..... 3
2.7. System Constraints ..... 4
2.8. Assumptions & Dependencies ..... 4
3. EXTERNAL INTERFACE REQUIREMENTS ..... 5
3.1. Hardware Interfaces ..... 5
3.2. Software Interfaces ..... 5
3.3. Communications Interfaces ..... 6
4. FUNCTIONAL REQUIREMENTS ..... 7
4.1. Functional Hierarchy ..... 7
4.2. Use Cases ..... 10
4.2.1. UC-01: Register Account ..... 11
4.2.2. UC-02: Verify Email ..... 12
4.2.3. UC-03: Log In ..... 13
4.2.4. UC-04: Manage Profile ..... 14
4.2.5. UC-05: Submit Hazard Report ..... 15
4.2.6. UC-06: Upload Queued Reports ..... 16
4.2.7. UC-07: Browse Hazard Map ..... 17
4.2.8. UC-08: Confirm Existing Hazard ..... 18
4.2.9. UC-09: Complete Volunteer Task ..... 19
4.2.10. UC-10: Submit Repair Check ..... 20
4.2.11. UC-11: Review Ranked Queue ..... 21
4.2.12. UC-12: Split Merged Pin ..... 22
4.2.13. UC-13: Decide Repair Claim ..... 23
4.2.14. UC-14: Override Status and Read Audit Log ..... 24
4.2.15. UC-15: Receive Weather Warning ..... 25
4.2.16. UC-16: View Points and Leaderboard ..... 26
4.2.17. UC-17: Claim Coupon ..... 27
4.2.18. UC-18: Redeem Coupon ..... 28
4.2.19. UC-19: Manage Coupons ..... 29
5. NON-FUNCTIONAL REQUIREMENTS ..... 30
5.1. Performance Requirements ..... 30
5.2. Safety Requirements ..... 30
5.3. Security Requirements ..... 30
5.4. User Documentation ..... 31
6. REFERENCES ..... 32
7. APPENDICES ..... 33

## List of Figures

Figure 1: System context diagram.....5
Figure 2: Functional hierarchy.....8
Figure 3: Use case diagram.....10

## List of Tables

Table 1: Intended audience.....1
Table 2: Terms, abbreviations and acronyms.....1
Table 3: Project objectives and measures.....3
Table 4: Stakeholders.....3
Table 5: Operating environment.....3
Table 6: Hardware interfaces.....5
Table 7: Software interfaces.....5
Table 8: Functional requirements.....8
Table 9: Performance requirements.....30
Table 10: Safety requirements.....30
Table 11: Security requirements.....30
Table 12: Traceability of objectives to requirements and use cases.....33
Table 13: Initial parameter values.....33

# 1. Introduction

## 1.1. Purpose of Document

This Software Requirements Specification defines the functions, interfaces and quality constraints of CivicPulse. It converts the approved proposal for project 23F-90 [12] into numbered requirements and use cases, against which the system can be built and assessed. The design that satisfies these requirements is described in the companion Software Design Specification.

## 1.2. Intended Audience

**Table 1: Intended audience**

| Reader | Use of this document |
|---|---|
| Supervisor and co-supervisor | Confirm that the requirements agree with the approved proposal and resolve open decisions. |
| FYP evaluation committee | Assess the scope and compare the delivered system with the stated requirements. |
| Development team | Plan work from individual requirements and derive tests from them. |
| Future maintainers | Understand the required behaviour before extending the system. |

## 1.3. Document Convention, abbreviations, and acronyms

Body text is set in Arial 11 point, headings in bold Arial and tables in Arial 9.5 to 10 point. Identifiers use fixed prefixes: FR for functional requirements, NFR for non-functional requirements, UC for use cases and O for project objectives. The word "shall" denotes a mandatory requirement. Bracketed numbers refer to the entries in Section 6.

**Table 2: Terms, abbreviations and acronyms**

| Term | Definition |
|---|---|
| Administrator / Reviewer | The person operating the dashboard. A team member performs this role during the demonstration. |
| API | Application Programming Interface |
| EXIF | Exchangeable Image File Format, metadata within an image that can include a position |
| Gate | One of five sequential checks applied to a claimed repair |
| GiST | Generalized Search Tree, the index type used for spatial queries |
| JWT | JSON Web Token, a signed token identifying a logged-in user |
| mAP@0.5 | Mean Average Precision at an overlap threshold of 0.5 |
| ORB | Oriented FAST and Rotated BRIEF, a keypoint detection and matching method [8] |
| OTP | One-Time Password, here the 6-digit email code |
| Pin | A single physical hazard on the map, grouping one or more reports |
| RANSAC | Random Sample Consensus, used to reject inconsistent keypoint matches |
| Report | One citizen submission comprising a photograph, category, position and accuracy value |
| SMTP | Simple Mail Transfer Protocol |
| Supporter | A distinct account that has reported or confirmed a pin |
| YOLO | You Only Look Once, a family of object detection models [9] |

# 2. Overall System Description

## 2.1. Project Background

Road damage in Sindh worsens rapidly during the monsoon. Provincial figures for the 2022 floods record 8,463 km of damaged road, including 1,055 km in Sukkur Division [1]. Available funding covers only part of this network; a Rs 1.5 billion allocation, for example, was limited to selected roads [2]. Municipal engineers must therefore prioritise repairs, which requires reliable information about each defect.

Existing channels provide little of this information. The Pakistan Citizen's Portal reports a closure rate of about 98.7%, although citizens mark a substantial share of those complaints as unresolved [3]. Helplines create one ticket for each caller. Platforms such as FixMyStreet limit duplicates through staff effort or by displaying existing reports [4]. Dashcam studies detect damage accurately, but require a survey vehicle and do not extend beyond detection [5]. An authority therefore cannot readily determine how many separate hazards exist, how severe they are or whether a repair has taken place.

To address these gaps, CivicPulse groups duplicate reports into a single pin, estimates severity from the photograph using a trained model, and accepts a repair only after photographic evidence has passed automatic checks and human review.

## 2.2. Project Scope

The project covers road hazards in Sukkur and delivers three connected products:

- **Android application:** hazard map, photographic reporting with an offline queue, hazard confirmation, volunteer tasks, repair checks, points, leaderboard and coupons.
- **Server:** report intake, secure image storage, merge engine, YOLO analysis in a background worker, five-gate repair pipeline, hazard ranking, weather warnings and reward rules.
- **Web dashboard:** ranked queue, map, pin detail and split function, claim review, coupon management, audit log and summary statistics.

The team will also collect and label three datasets: local road photographs, a ground-truth set for merge evaluation and a set of deliberately faked repairs.

## 2.3. Not in Scope

- Physical road repair, and any volunteer activity that would endanger a person, including pothole filling in live traffic.
- Live municipal deployment. No agreement exists with Sukkur Municipal Corporation, so a team member will act as reviewer and results will be reported accordingly.
- A contracted shop network. If no shop participates, coupons will be simulated and labelled as such.
- Ranking weights learned from data, since no labelled ranking dataset is available.
- Detection of one person operating several verified accounts. The per-pin cap and human gate limit the impact of this abuse but do not detect it.
- Formal security auditing, navigation and prediction of future defects.
- An iPhone build and cloud hosting, which remain optional after the defence.

## 2.4. Project Objectives

Table 3 lists the project objectives. Each has a defined measure, and all except the stretch objective O7 must be achieved for the defence.

**Table 3: Project objectives and measures**

| ID | Objective | Measure |
|---|---|---|
| O1 | Merge reports of one physical hazard into one pin using distance, category and time. | Precision and recall against a hand-labelled set, with a radius sweep from 5 m to 40 m using ST_ClusterDBSCAN [7]. |
| O2 | Detect road-surface defects and estimate severity from a single photograph. | mAP@0.5 on unseen Sukkur photographs, for the public-data model and the fine-tuned model. |
| O3 | Verify a claimed repair from a before-and-after pair of photographs. | False-accept and false-reject rates on faked and genuine repairs, with gates disabled and then enabled. |
| O4 | Rank hazards using a transparent score. | Spearman's rho against independent rankings of about 50 hazards by three people. |
| O5 | Warn users in an affected zone and raise the priority of weather-sensitive hazards. | Median delivery time and the proportion of notified users inside the zone. |
| O6 | Award points, order the leaderboard and issue single-use coupons correctly. | An automated test suite, including simultaneous redemption of one coupon. |
| O7 (stretch) | Observe real use, subject to recruiting a user group. | Indicative figures from a two-week trial. |

## 2.5. Stakeholders

**Table 4: Stakeholders**

| Stakeholder | Role or interest |
|---|---|
| Citizens and volunteers | Report and confirm hazards, carry out safe tasks and earn rewards. |
| Municipal reviewer (performed by a team member) | Receives a ranked list with supporting evidence and approves repairs. |
| Sukkur Municipal Corporation (potential) | May obtain a measured record of hazards and repairs if the system is adopted. |
| Local shops (potential) | May provide coupons in return for visibility. |
| Ayoush Kumar | Detector, severity estimation, Gates 3 and 4, ground-truth and faked-repair datasets. |
| Sarichand | Accounts, reporting, image handling, Gates 1 and 2, ranking, weather service and radius sweep. |
| Rainuka Bai | Map, dashboard, audit log, merge engine, points, leaderboard and coupons. |
| Dr. Saif Hassan and Dr. Ismail | Project supervision and approval of key decisions. |

## 2.6. Operating Environment

**Table 5: Operating environment**

| Component | Environment |
|---|---|
| Mobile application | Android phones with a camera and GPS, running a React Native build created with Expo. The application shall remain usable with weak or no coverage. |
| Dashboard | A current desktop web browser on a large screen. |
| Server | A laptop or small server running Docker Compose, with Python 3.12, FastAPI, PostgreSQL 16 and PostGIS 3.4. YOLO runs on the CPU for the demonstration. |
| Model training | Free GPU sessions on Google Colab or Kaggle, separate from the operational system. |

| Component | Environment |
|---|---|
| Network | Mobile data or Wi-Fi between clients and server. Internet access is required for forecasts, push messages and map tiles. |

## 2.7. System Constraints

- **Software:** only free and open tools may be used. The application is limited to the capabilities of Expo; remote push and the map library require a development build instead of Expo Go.
- **Hardware:** no hardware will be purchased. Inference must run on an ordinary laptop, which favours a small YOLO variant.
- **Cultural:** this version uses English, and public views show a chosen display name in place of a real name.
- **Legal:** photographs and positions are personal data, so their exposure is minimised. Faces and number plates in collected datasets are blurred, and volunteer tasks must be lawful for private persons.
- **Environmental:** use is outdoors, in sunlight and rain, near moving traffic. Screens therefore need large controls and few steps, and the application must not require operation while driving.
- **User:** users have varied technical ability and may own low-end phones, so workflows must be short and error messages clear.
- **Off-the-shelf components:** Open-Meteo terms apply to non-commercial use [10], public map tiles require attribution, and phone GPS is typically accurate to about 4.9 m in open sky and less accurate near buildings [6].
- **Academic:** the work must fit the 26-week plan and be demonstrable at the defence.

## 2.8. Assumptions & Dependencies

**Assumptions.** The following working decisions apply until confirmed by the supervisor:

- Verification uses an emailed code, as SMS incurs a cost. The proposal permits either a phone number or an email address.
- Hazard categories are pothole, waterlogged road, blocked drain, fallen branch or debris, and other. The detector covers road-surface defects only; other categories use a default severity and administrator review.
- Merge parameters, score weights, point values and weather thresholds in Appendix B are initial values, to be tuned using measured data.
- A team member acts as municipal reviewer, and also as the shop when coupons are simulated.

**Dependencies.** The system depends on Open-Meteo, the Expo push service, an SMTP provider and OpenStreetMap tiles. It also depends on the accuracy value and mock-location flag reported by the phone [11], the RDD2022 dataset [5] and timely collection of local data by the team.

# 3. External Interface Requirements

CivicPulse exchanges data with its users, the phone hardware and four external services, as shown in the context diagram below. Clients communicate only with the server API, except when downloading map tiles.

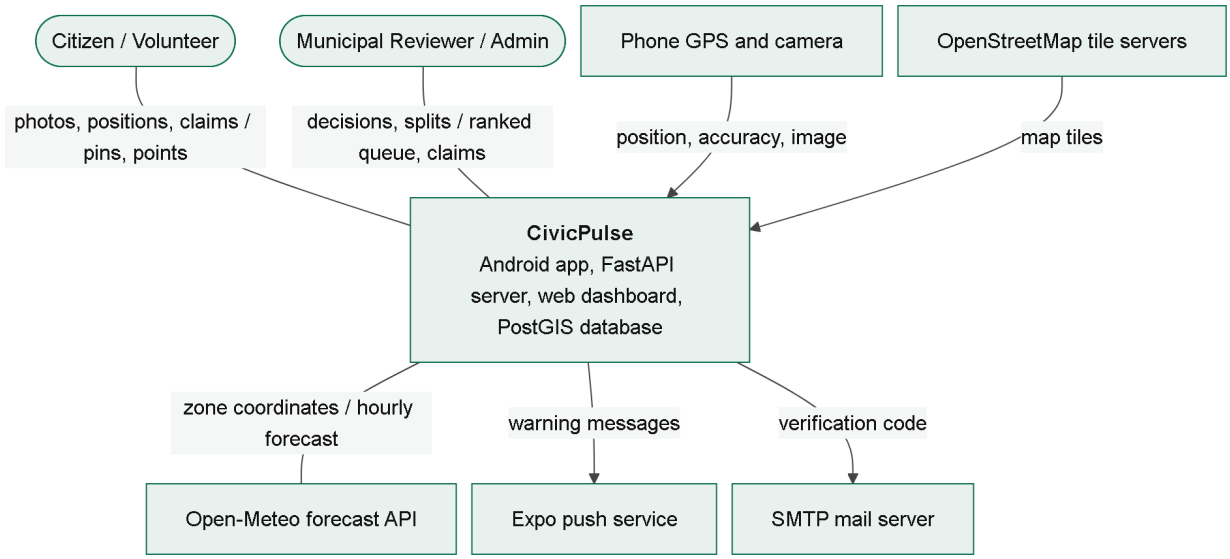



**Figure 1: System context diagram**

## 3.1. Hardware Interfaces

**Table 6: Hardware interfaces**

| Device | Interaction |
|---|---|
| Phone camera | Captures photographs in the application through expo-camera. Gallery selection is not offered. |
| Phone GPS | Supplies latitude, longitude, accuracy in metres and a mock-location flag through expo-location, only while the application is open. |
| Phone storage | Holds queued reports and photographs until upload, and tokens in the secure keystore. |
| Server disk | Stores re-encoded photographs under random file names. |
| Desktop or laptop | Runs the dashboard in a web browser. |

## 3.2. Software Interfaces

**Table 7: Software interfaces**

| Software | Version | Purpose and data exchanged |
|---|---|---|
| PostgreSQL + PostGIS | 16 / 3.4 | Stores all records. Geography columns and spatial indexes answer distance queries. |
| FastAPI, SQLAlchemy, Alembic | 0.162 / 2.1 / 1.20 | Web API, database access and schema migrations. |
| React Native + Expo | SDK 58 | Mobile runtime, secure storage, camera, location and push registration. |
| React + Vite | Current stable | Dashboard front end. |
| MapLibre + OpenStreetMap | Current stable | Map display on both clients, with tiles downloaded per viewport. |

| Software | Version | Purpose and data exchanged |
|---|---|---|
| Ultralytics YOLO [9] | Fixed in Phase 4 | Receives an image and returns boxes, classes and confidence values. |
| OpenCV | Fixed in Phase 5 | ORB features, matching and homography estimation for Gate 3. |
| Pillow | Current stable | Decodes and re-encodes uploads and removes metadata. |
| Open-Meteo API [10] | Public service | Receives zone coordinates and returns hourly rainfall, rain probability and wind gusts. |
| Expo push service | Public service | Receives device tokens and warning text and delivers the notification. |

## 3.3. Communications Interfaces

- **Client to server:** REST over HTTP, with HTTPS required on any public address. A local demonstration may use HTTP on a private network. Request bodies use JSON, except photograph uploads, which use multipart form data.
- **Authentication:** a bearer JWT on each protected request, and an X-Device-Id header identifying the phone for rate limiting.
- **Status codes:** 400 incorrect code, 401 not authenticated, 403 wrong role or unverified account, 409 conflict, 422 invalid input, 423 locked code, and 429 rate limited with a Retry-After header.
- **Email:** SMTP, with STARTTLS where the provider supports it. Development email is captured by a local Mailpit inbox.
- **Synchronisation:** the application retries uploads with increasing delays, and the server treats a repeated client identifier as the same report.
- **Media:** photographs are served from an origin separate from the dashboard, with content sniffing disabled.

# 4. Functional Requirements

## 4.1. Functional Hierarchy

The system comprises nine functional areas, shown in the hierarchy below and specified in the requirements table that follows. Priority is Must for functions required at the defence, and Should for functions that may be reduced under the project's risk plan.

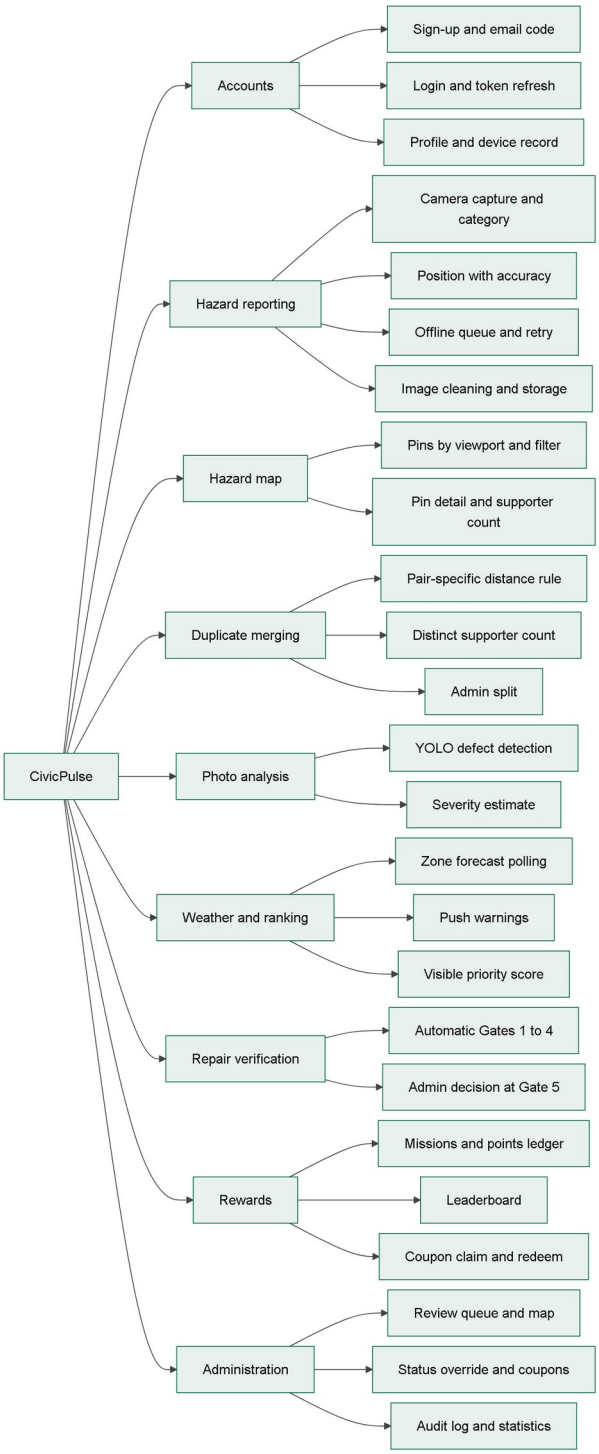



**Figure 2: Functional hierarchy**

**Table 8: Functional requirements**

| ID | Requirement: the system shall | Priority |
|---|---|---|
| FR-01 | Register a citizen with email, password and display name, and reject duplicate emails. | Must |
| FR-02 | Verify an email using a 6-digit code that expires in 10 minutes and locks after 5 incorrect attempts. | Must |

| ID | Requirement: the system shall | Priority |
|---|---|---|
| FR-03 | Issue a 15-minute access token and a 14-day refresh token at login. | Must |
| FR-04 | Refuse protected actions from unverified or disabled accounts, and administrator actions from citizens. | Must |
| FR-05 | Create administrator accounts only through a server-side script, never through the API. | Must |
| FR-06 | Allow users to change their display name and register a device. | Must |
| FR-07 | Capture a report containing an in-app photograph, category, optional note, position, accuracy, time and mock-location flag. | Must |
| FR-08 | Queue reports on the phone when offline and upload each of them exactly once. | Must |
| FR-09 | Verify uploads by content, limit their size, re-encode them, remove metadata, and store them under random names with a SHA-256 hash. | Must |
| FR-10 | Display open pins for the visible map area with filters and supporter counts, and without reporter identity. | Must |
| FR-11 | Attach a report to the nearest open pin of the same category inside the pair threshold and active window, and otherwise create a new pin. | Must |
| FR-12 | Count supporters as distinct accounts and record each merge decision on its report. | Must |
| FR-13 | Never add a report to a closed pin; instead, link a new pin to the closed one. | Must |
| FR-14 | Allow an administrator to split a pin with no loss of reports. | Must |
| FR-15 | Analyse each photograph with YOLO in the background, and store the boxes and model version. | Must |
| FR-16 | Compute severity from 1 to 5, using a category default when nothing is detected. | Must |
| FR-17 | Poll hourly forecasts for each zone, retain snapshots and continue operating after a failed call. | Must |
| FR-18 | Send one push warning per event to devices in the affected zone, and record sent and received times. | Must |
| FR-19 | Compute a priority score with a visible breakdown, and recompute it when its inputs change. | Must |
| FR-20 | Store ranking weights as versioned settings and log each change. | Should |
| FR-21 | Issue a one-time token when a repair mission is accepted, and refuse a checker who reported the pin. | Must |
| FR-22 | Run Gates 1 to 4 in order, stop at the first failure and store all measurements. | Must |
| FR-23 | Pay no reward before administrator approval at Gate 5, and allow approval of an automatically rejected claim. | Must |
| FR-24 | Provide switches that disable Gates 2, 3 and 4 for the evaluation run. | Must |
| FR-25 | Record points in an append-only ledger with a unique award key, and accept no point value from a client. | Must |
| FR-26 | Hold report points until the pin has a second supporter or administrator approval, and enforce a per-pin cap. | Must |
| FR-27 | Order the leaderboard by points using a stable tie rule. | Must |
| FR-28 | Issue coupon codes against a balance, and redeem each code at most once, including under simultaneous requests. | Must |
| FR-29 | Offer safe volunteer tasks, verified by before-and-after photographs and administrator approval. | Should |
| FR-30 | Provide a ranked queue, map, pin detail, claim review, coupon management and statistics on the dashboard. | Must |
| FR-31 | Write an audit record for each approval, rejection, override, merge correction and split, in the same transaction as the change. | Must |

## 4.2. Use Cases

The use case diagram below shows the primary and secondary actors and their use cases. A volunteer is a citizen who has accepted a mission, and therefore inherits all citizen use cases. External services participate as secondary actors.

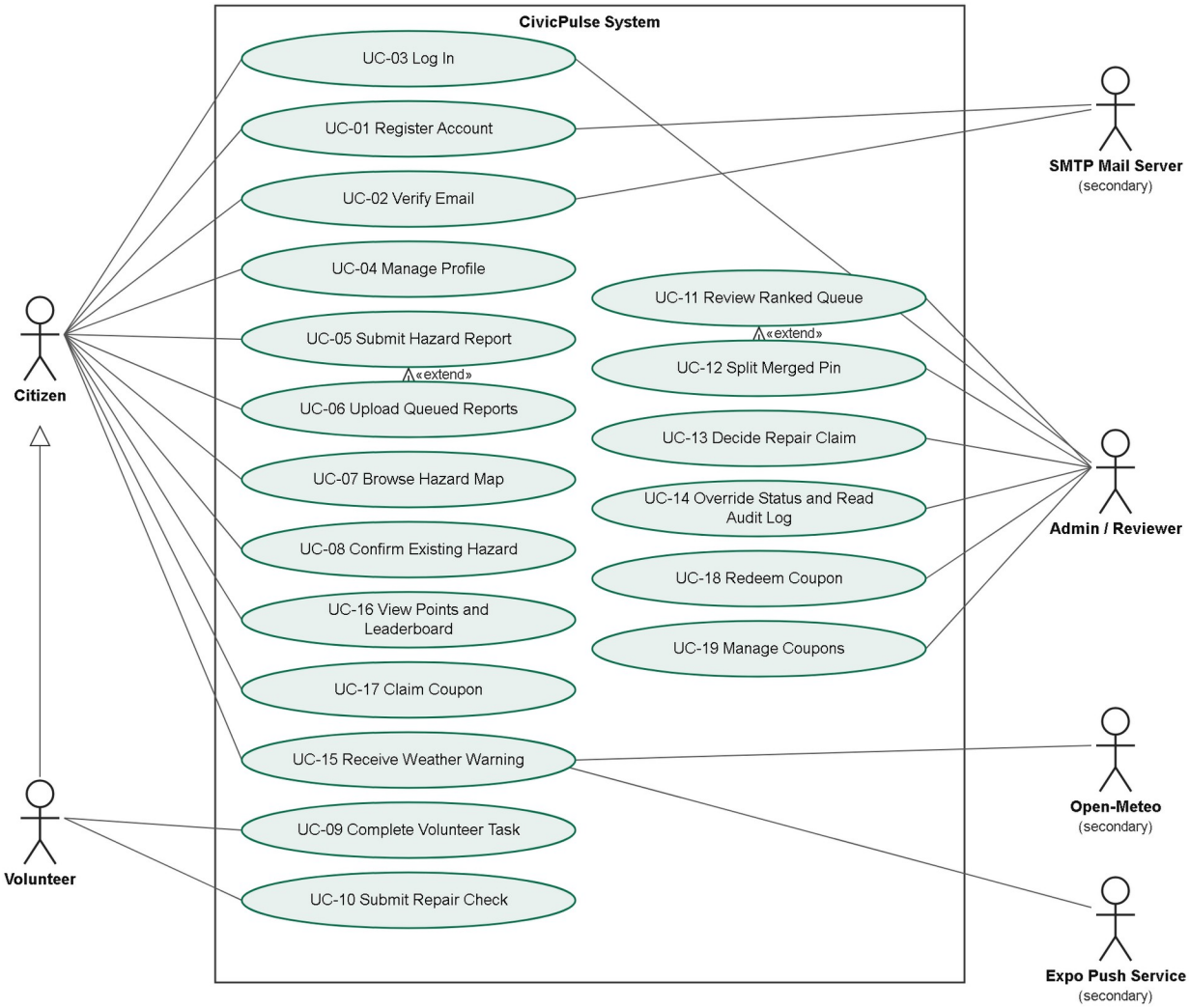



**Figure 3: Use case diagram**

### 4.2.1. UC-01: Register Account

A new user creates a citizen account, so that subsequent reports and rewards can be attributed to one person.

| **UC-01: Register Account** | | |
|---|---|---|
| **Use case Id:** | UC-01 | |
| **Actors:** Citizen (initiator), SMTP mail server | | |
| **Feature:** Accounts | | |
| **Pre-condition:** | The app is installed and the phone can reach the server. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the sign-up screen and enters a display name, email and password. | Checks the format of each field on the phone. |
| 2. | Taps Create account. | Counts the request against the device and network limits, hashes the password with Argon2 and saves the account as unverified. |
| 3. | (System step) | Generates a 6-digit code, stores only its keyed hash, emails the code and opens the code entry screen. |
| **Alternate Scenarios** | | |
| 2a: The email already belongs to an account. The server answers 409 and the app suggests logging in.<br>2b: The password has fewer than 8 characters. The server answers 422 and nothing is stored.<br>2c: Too many sign-ups came from this device. The server answers 429 with a retry time. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | An unverified citizen account exists. | |
| 2. | One active verification code exists for that account. | |
| **Use Case Cross referenced** | UC-02 | |

### 4.2.2. UC-02: Verify Email

The citizen confirms ownership of the email address by entering the code sent to it.

| **UC-02: Verify Email** | | |
|---|---|---|
| **Use case Id:** | UC-02 | |
| **Actors:** Citizen | | |
| **Feature:** Accounts | | |
| **Pre-condition:** | An unverified account exists and a code has been sent. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Types the 6-digit code and taps Verify. | Locks the code record, then checks its expiry and attempt count. |
| 2. | (System step) | Compares the code, marks it used and marks the account verified. |
| 3. | (System step) | Returns the citizen to the login screen with a confirmation message. |
| **Alternate Scenarios** | | |
| 1a: The code is wrong. The attempt count rises and the citizen may try again.<br>1b: This is the fifth wrong try. The code locks and even the right value is refused (423).<br>1c: Ten minutes have passed. The server reports an expired code.<br>1d: The citizen taps Send a new code. The old code is cancelled and a new one is emailed, at most three times in ten minutes. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The account is verified and can log in. | |
| **Use Case Cross referenced** | UC-01, UC-03 | |

### 4.2.3. UC-03: Log In

A verified citizen or an administrator obtains access and refresh tokens. The phone retains the tokens across application restarts.

| **UC-03: Log In** | | |
|---|---|---|
| **Use case Id:** | UC-03 | |
| **Actors:** Citizen or Admin | | |
| **Feature:** Accounts | | |
| **Pre-condition:** | The account exists, is verified and is not disabled. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Enters email and password and taps Log in. | Counts the attempt per account, per device and per network address. |
| 2. | (System step) | Verifies the Argon2 hash and issues a 15-minute access token and a 14-day refresh token with the profile. |
| 3. | (System step) | The app stores the tokens in SecureStore, registers the device and opens the main screen. |
| **Alternate Scenarios** | | |
| 1a: The credentials are wrong. A single generic message is shown, so nobody learns whether the email exists.<br>1b: The email is unverified. The app moves to the code entry screen.<br>1c: A sixth attempt within a minute receives 429.<br>3a: The access token expires later. The app silently requests a new one with the refresh token. If that is rejected, the citizen returns to login. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The client holds valid tokens. | |
| 2. | The device record is current. | |
| **Use Case Cross referenced** | UC-02, UC-04 | |

### 4.2.4. UC-04: Manage Profile

The citizen views the account details and changes the name displayed on the leaderboard.

| **UC-04: Manage Profile** | | |
|---|---|---|
| **Use case Id:** | UC-04 | |
| **Actors:** Citizen | | |
| **Feature:** Accounts | | |
| **Pre-condition:** | The citizen is logged in. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the profile screen. | Shows email, role and display name. |
| 2. | Edits the display name and taps Save. | Validates the length, saves the new name and returns the updated profile. |
| **Alternate Scenarios** | | |
| 2a: The name is shorter than 2 or longer than 40 characters. The server answers 422.<br>2b: A request also tries to set a role. The field is ignored and the role does not change. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The new display name is used wherever the citizen is shown. | |
| **Use Case Cross referenced** | UC-03, UC-16 | |

### 4.2.5. UC-05: Submit Hazard Report

The citizen photographs a hazard on site and submits it with a category and position. This is the principal citizen function.

| **UC-05: Submit Hazard Report** | | |
|---|---|---|
| **Use case Id:** | UC-05 | |
| **Actors:** Citizen | | |
| **Feature:** Hazard reporting | | |
| **Pre-condition:** | The citizen is logged in and has granted camera and foreground location permission. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the report screen and takes a photo with the in-app camera. | Records latitude, longitude, accuracy radius, capture time and the mock-location flag. |
| 2. | Picks a category, optionally adds a note and taps Send. | Saves the report on the phone with a client-generated identifier and starts the upload. |
| 3. | (System step) | The server cleans and stores the photo, then applies the merge rule. |
| 4. | (System step) | The report is attached to a pin, a vision job is queued and the app shows the report status. |
| **Alternate Scenarios** | | |
| 1a: A permission was refused. The app explains why it is needed and how to allow it.<br>2a: There is no signal. The report stays queued and UC-06 takes over.<br>3a: The file is not a real image or exceeds 15 MB. The upload is rejected with a clear reason.<br>3b: The position is flagged as mocked. The report is kept but marked as untrusted. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The report and its image are stored permanently. | |
| 2. | The report belongs to exactly one pin. | |
| 3. | The pin's supporter count reflects distinct accounts. | |
| **Use Case Cross referenced** | UC-06, UC-07, UC-08 | |

### 4.2.6. UC-06: Upload Queued Reports

Reports created without network coverage are delivered automatically once a connection returns, and each is stored exactly once.

| **UC-06: Upload Queued Reports** | | |
|---|---|---|
| **Use case Id:** | UC-06 | |
| **Actors:** Citizen (indirectly), Mobile app | | |
| **Feature:** Hazard reporting | | |
| **Pre-condition:** | At least one report is waiting in the phone's local queue. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Regains a network connection or reopens the app. | Detects connectivity and sends the oldest queued report. |
| 2. | (System step) | The server checks the client identifier and stores the report if it is new. |
| 3. | (System step) | After the confirmation arrives, the app deletes the item from its queue and updates My Reports. |
| **Alternate Scenarios** | | |
| 1a: The upload fails. The app retries later with increasing delay and keeps the item.<br>2a: The server already holds this identifier, because an earlier confirmation was lost. It returns the existing report and creates no duplicate. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | Each queued report exists once on the server. | |
| 2. | The local queue no longer holds confirmed items. | |
| **Use Case Cross referenced** | UC-05 | |

### 4.2.7. UC-07: Browse Hazard Map

A logged-in user views the hazards in the visible map area. Reporter identities are not disclosed.

| **UC-07: Browse Hazard Map** | | |
|---|---|---|
| **Use case Id:** | UC-07 | |
| **Actors:** Citizen | | |
| **Feature:** Hazard map | | |
| **Pre-condition:** | The citizen is logged in. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the map or moves it. | Requests pins inside the visible area and draws them with supporter badges. |
| 2. | Applies a category or status filter. | Requests and draws the matching pins. |
| 3. | Taps a pin. | Shows photo, category, severity, status and supporter count. |
| **Alternate Scenarios** | | |
| 1a: There is no connection. The app shows the last loaded pins with a notice.<br>1b: Closed pins are left out by default. A new pin at the same place links to the closed one it follows. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | No data has changed. | |
| **Use Case Cross referenced** | UC-05, UC-08, UC-15 | |

### 4.2.8. UC-08: Confirm Existing Hazard

A second person adds supporting evidence to a hazard reported by someone else, which increases its corroboration count.

| **UC-08: Confirm Existing Hazard** | | |
|---|---|---|
| **Use case Id:** | UC-08 | |
| **Actors:** Citizen | | |
| **Feature:** Missions and rewards | | |
| **Pre-condition:** | The citizen is logged in, stands near an open pin and did not report it. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Taps Confirm on the pin detail sheet. | Opens the camera for a fresh photo. |
| 2. | Takes a photo and sends it. | Checks the position against the same pair-specific threshold used for merging. |
| 3. | (System step) | Attaches the photo as a new report, recounts distinct supporters and awards the confirmation points once. |
| **Alternate Scenarios** | | |
| 2a: The citizen is too far away. The confirmation is refused and no points are awarded.<br>3a: The citizen reported or already confirmed this pin. The evidence may be kept but pays nothing. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The supporter count is correct. | |
| 2. | The original reporter's held points may now be released. | |
| **Use Case Cross referenced** | UC-05, UC-07, UC-16 | |

### 4.2.9. UC-09: Complete Volunteer Task

A volunteer carries out a lawful and safe task, such as clearing debris or unblocking a drain. Pothole filling is excluded.

| **UC-09: Complete Volunteer Task** | | |
|---|---|---|
| **Use case Id:** | UC-09 | |
| **Actors:** Volunteer, Admin | | |
| **Feature:** Missions and rewards | | |
| **Pre-condition:** | The volunteer is logged in and a task is available on an open pin. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the mission list and accepts a task. | Records the mission and shows safety guidance. |
| 2. | Takes before and after photos on site and submits them. | Applies the freshness and position checks. |
| 3. | (System step) | Places the task in the admin's review list. |
| 4. | (System step) | When the admin approves, awards the task points once. |
| **Alternate Scenarios** | | |
| 2a: A check fails. The volunteer sees the reason and no points are awarded.<br>4a: The admin rejects the task. The reason is logged and shown to the volunteer. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The mission has a final status. | |
| 2. | Any award appears once in the ledger. | |
| **Use Case Cross referenced** | UC-13, UC-16 | |

## 4.2.10. UC-10: Submit Repair Check

A volunteer photographs a location that appears to have been repaired, and the server evaluates the photograph through the automatic gates in order of cost.

| **UC-10: Submit Repair Check** | | |
|---|---|---|
| **Use case Id:** | UC-10 | |
| **Actors:** Volunteer | | |
| **Feature:** Repair verification | | |
| **Pre-condition:** | The pin is open or pending repair, and the volunteer is not its reporter. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Accepts the repair-check mission while online. | Issues a one-time token with an expiry and stores only its hash. |
| 2. | Reads the framing tips and takes the after photo. | Queues the photo, so it can upload later if coverage is lost. |
| 3. | (System step) | Gate 1: confirms the token and that the image hash is not already on this pin. |
| 4. | (System step) | Gate 2: confirms the position lies inside the pin's threshold and notes any mock flag. |
| 5. | (System step) | Gate 3: matches ORB keypoints with the before photo and requires enough RANSAC inliers. |
| 6. | (System step) | Gate 4: maps the defect box into the after photo and runs YOLO on that patch only. |
| 7. | (System step) | Saves all measurements and marks the claim as awaiting the admin. |
| **Alternate Scenarios** | | |
| 1a: The volunteer is the pin's reporter. The mission is refused.<br>3a to 6a: A gate fails. The run stops there, later gates are skipped, and the claim records the failure. The admin may still review it.<br>6b: The mapped box falls outside the frame or is too small. This counts as a Gate 4 failure. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | A repair claim exists with every gate result. | |
| 2. | No points have been paid. | |
| **Use Case Cross referenced** | UC-13, UC-16 | |

## 4.2.11. UC-11: Review Ranked Queue

The reviewer examines the ranked list of open hazards and the contribution of each factor to a hazard's position.

| **UC-11: Review Ranked Queue** | | |
|---|---|---|
| **Use case Id:** | UC-11 | |
| **Actors:** Admin | | |
| **Feature:** Administration | | |
| **Pre-condition:** | The admin is logged in to the dashboard. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the queue. | Lists open pins by priority score, with ties broken in favour of the older pin. |
| 2. | Selects a pin. | Shows severity, supporters, days open, weather risk and each one's share of the score. |
| 3. | Opens the pin detail. | Shows the reports, photos and detection boxes. |
| **Alternate Scenarios** | | |
| 1a: A citizen token calls this route. The server answers 403.<br>2a: A photo could not be analysed. The pin is flagged and uses its category's default severity. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | No data has changed unless the admin continues to UC-12 or UC-14. | |
| **Use Case Cross referenced** | UC-12, UC-13, UC-14 | |

## 4.2.12. UC-12: Split Merged Pin

An administrator corrects an incorrect automatic merge. No data is lost, because reports remain separate records attached to a pin.

| **UC-12: Split Merged Pin** | | |
|---|---|---|
| **Use case Id:** | UC-12 | |
| **Actors:** Admin | | |
| **Feature:** Duplicate merging | | |
| **Pre-condition:** | The pin has two or more reports. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the pin and selects the reports that belong elsewhere. | Shows each report's merge decision: candidates, distances and thresholds. |
| 2. | Confirms the split. | Creates a second pin in one transaction and moves the selected reports. |
| 3. | (System step) | Recounts supporters on both pins, recomputes scores and writes an audit row. |
| **Alternate Scenarios** | | |
| 2a: All reports or none are selected. The split is refused. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | Two pins exist and every original report is intact. | |
| 2. | The audit log names the admin, the time and both states. | |
| **Use Case Cross referenced** | UC-11, UC-14 | |

## 4.2.13. UC-13: Decide Repair Claim

This use case implements Gate 5. A human decision is required because cooperating accounts could satisfy every automatic check.

| **UC-13: Decide Repair Claim** | | |
|---|---|---|
| **Use case Id:** | UC-13 | |
| **Actors:** Admin | | |
| **Feature:** Repair verification | | |
| **Pre-condition:** | A claim is awaiting a decision. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the claim list and selects a claim. | Shows before and after photos, matched keypoints and each gate's measurements. |
| 2. | Approves the claim, with an optional note. | Closes the pin and writes an audit row in the same transaction. |
| 3. | (System step) | Awards the repair points once, within the per-pin cap, and notifies the volunteer. |
| **Alternate Scenarios** | | |
| 2a: The admin rejects the claim. The pin stays open, no points are paid and the reason is logged.<br>2b: The admin approves a claim that an automatic gate rejected. This is allowed and is logged as an override.<br>3a: The cap has been reached. The claim closes and no further points are paid. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The claim has a final decision. | |
| 2. | Points are in the ledger only if the claim was approved. | |
| **Use Case Cross referenced** | UC-10, UC-16 | |

## 4.2.14. UC-14: Override Status and Read Audit Log

An administrator manually corrects the status of a pin, and can later review all sensitive actions in the audit log.

| **UC-14: Override Status and Read Audit Log** | | |
|---|---|---|
| **Use case Id:** | UC-14 | |
| **Actors:** Admin | | |
| **Feature:** Administration | | |
| **Pre-condition:** | The admin is logged in. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Selects a new status for a pin and writes a note. | Saves the change and an audit row in one transaction. |
| 2. | Opens the audit log and applies filters. | Lists actor, action, target, before and after values, and time. |
| **Alternate Scenarios** | | |
| 1a: The status value is not allowed. The server answers 422 and nothing changes. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The audit log holds the override. | |
| 2. | Audit rows are never edited through the system. | |
| **Use Case Cross referenced** | UC-11, UC-12, UC-13 | |

## 4.2.15. UC-15: Receive Weather Warning

Citizens in a zone are warned before forecast rain makes road damage harder to see, and rain-sensitive hazards are ranked higher.

| **UC-15: Receive Weather Warning** | | |
|---|---|---|
| **Use case Id:** | UC-15 | |
| **Actors:** Open-Meteo, Expo push service, Citizen | | |
| **Feature:** Weather and ranking | | |
| **Pre-condition:** | Zones are defined and the phone has shared its current zone while the app was open. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | (System step) | The scheduler fetches the forecast for each zone every hour and stores a snapshot. |
| 2. | (System step) | When the risk value crosses the threshold, it logs the crossing time and recomputes scores in that zone. |
| 3. | (System step) | Sends one push message to devices whose last known zone matches. |
| 4. | Sees the warning on the phone. | The app reports the arrival time so that delivery delay can be measured. |
| **Alternate Scenarios** | | |
| 1a: Open-Meteo cannot be reached. The last snapshot stays in use and the failure is logged.<br>2a: A warning for the same event has already gone out. No repeat is sent. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | A notification record holds the sent and received times. | |
| 2. | Pin scores reflect the new risk. | |
| **Use Case Cross referenced** | UC-07, UC-11 | |

## 4.2.16. UC-16: View Points and Leaderboard

The citizen views a balance, its history and the leaderboard. The server calculates all values.

| **UC-16: View Points and Leaderboard** | | |
|---|---|---|
| **Use case Id:** | UC-16 | |
| **Actors:** Citizen | | |
| **Feature:** Missions and rewards | | |
| **Pre-condition:** | The citizen is logged in. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the points screen. | Sums the ledger and returns the balance with its history. |
| 2. | Opens the leaderboard. | Orders users by points. A tie goes to whoever reached the total first. |
| **Alternate Scenarios** | | |
| 1a: Report points are still held. They are shown as pending until the pin is corroborated or approved. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | No data has changed. | |
| **Use Case Cross referenced** | UC-08, UC-10, UC-13, UC-17 | |

## 4.2.17. UC-17: Claim Coupon

The citizen exchanges points for a coupon and receives a unique code.

| **UC-17: Claim Coupon** | | |
|---|---|---|
| **Use case Id:** | UC-17 | |
| **Actors:** Citizen | | |
| **Feature:** Missions and rewards | | |
| **Pre-condition:** | The citizen is logged in and an active coupon is in stock. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Opens the coupon list and chooses one. | Shows the cost, the shop and whether the coupon is simulated. |
| 2. | Taps Claim. | Locks the user row, checks the balance and stock, then writes a spending entry and an issued code in one transaction. |
| **Alternate Scenarios** | | |
| 2a: The balance or the stock is too low. The server answers 409 and nothing changes. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The balance is lower by the cost. | |
| 2. | An issued code belongs to the citizen. | |
| **Use Case Cross referenced** | UC-16, UC-18 | |

## 4.2.18. UC-18: Redeem Coupon

An issued code is redeemed once. In the absence of a shop partner, an administrator screen performs this role.

| **UC-18: Redeem Coupon** | | |
|---|---|---|
| **Use case Id:** | UC-18 | |
| **Actors:** Admin (acting as shop) | | |
| **Feature:** Missions and rewards | | |
| **Pre-condition:** | The citizen holds an issued code. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Enters the code. | Runs one conditional update that succeeds only if the code is still issued. |
| 2. | (System step) | Confirms the redemption and records the time. |
| **Alternate Scenarios** | | |
| 1a: Two devices try at the same moment. Exactly one succeeds and the other receives 409.<br>1b: The code does not exist. The server reports that it was not found. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The code is permanently redeemed. | |
| **Use Case Cross referenced** | UC-17, UC-19 | |

## 4.2.19. UC-19: Manage Coupons

An administrator maintains the coupon catalogue.

| **UC-19: Manage Coupons** | | |
|---|---|---|
| **Use case Id:** | UC-19 | |
| **Actors:** Admin | | |
| **Feature:** Administration | | |
| **Pre-condition:** | The admin is logged in. | |
| **Scenarios** | | |
| **Step#** | **Action** | **Software Reaction** |
| 1. | Enters the title, shop, cost, stock and simulated flag. | Validates and saves the coupon and writes an audit row. |
| 2. | Deactivates a coupon. | Hides it from the catalogue. Codes already issued stay valid. |
| **Alternate Scenarios** | | |
| 1a: The cost or stock is negative. The server answers 422. | | |
| **Post Conditions** | | |
| **Step#** | **Description** | |
| 1. | The catalogue matches the admin's changes. | |
| 2. | The change appears in the audit log. | |
| **Use Case Cross referenced** | UC-17, UC-18 | |

# 5. Non-functional Requirements

## 5.1. Performance Requirements

The approved proposal does not set pass or fail thresholds for performance. The values in Table 9 are therefore design targets, to be measured during testing. They are not results that have already been achieved.

**Table 9: Performance requirements**

| ID | Requirement |
|---|---|
| NFR-01 | A normal API request, including the merge lookup, should complete within 1 second on the demonstration laptop using the seeded dataset. |
| NFR-02 | The merge lookup shall use the spatial index and shall not scan all stored reports. |
| NFR-03 | YOLO inference shall not run within a user request, and should take about 2 seconds per image on the CPU. |
| NFR-04 | A confirmed report shall survive application restarts and repeated uploads without loss or duplication. |
| NFR-05 | Forecasts shall refresh hourly, and warning delay shall be reported as a measured median. |
| NFR-06 | Uploads shall be limited to 15 MB, and stored photographs to 1600 pixels on the long side. |
| NFR-07 | The complete server stack shall start with a single command. |

## 5.2. Safety Requirements

**Table 10: Safety requirements**

| ID | Requirement |
|---|---|
| NFR-08 | The application shall offer only tasks that a private person can lawfully and safely perform, and shall display safety guidance before a mission starts. |
| NFR-09 | The application shall not encourage use while driving. Reporting requires a deliberate photograph, and warnings use brief text. |
| NFR-10 | An incorrect merge shall be reversible, and merging shall never delete an original report. |
| NFR-11 | No reward shall result from an automatic decision alone, so a software fault cannot release a payout without human approval. |
| NFR-12 | A failure of the model or the weather service shall affect only that function, and reports shall still be accepted. |

No safety certification applies to this academic prototype.

## 5.3. Security Requirements

**Table 11: Security requirements**

| ID | Requirement |
|---|---|
| NFR-13 | The server shall be the sole authority. Client values are treated as claims, and no point value is accepted from a client. |
| NFR-14 | Passwords shall be hashed with Argon2, and verification codes and mission tokens stored only as hashes. |
| NFR-15 | Rate limits shall apply per account and per device; for example, a sixth login attempt within one minute receives status 429. |
| NFR-16 | Exact coordinates shall be available only to the reporter and administrators, and public views shall not identify a reporter. |
| NFR-17 | Location shall be read only while the application is open, and a phone shall share only its zone identifier for weather targeting. |

| ID | Requirement |
|---|---|
| NFR-18 | Uploads shall be treated as data: the type is determined from file content, and files are re-encoded and served from a separate origin. |
| NFR-19 | User text shall be escaped on output, and the dashboard shall not render raw HTML from stored fields. |
| NFR-20 | Secrets and raw photographs shall not be committed to the source repository. |
| NFR-21 | Each sensitive administrator action shall be traceable to a person and a time. |

## 5.4. User Documentation

- A citizen guide covering registration, reporting, confirmation and coupons.
- A reviewer guide covering the queue, pin splitting, claim review and the audit log.
- Setup instructions in the repository, and API documentation generated at the /docs address.
- An evaluation guide explaining how the figures for objectives O1 to O6 are regenerated.

# 6. References

[1] Provincial Disaster Management Authority, Government of Sindh, Flood 2022 in Sindh, from June 20, 2022 to March 31, 2023, p. 17. [Online]. Available: https://pdma.gos.pk/Documents/Reports/Flood%202022%20In%20Sindh.pdf (accessed Sep. 3, 2026).

[2] "Murad allocates Rs1.5bn for repair of few select rain-damaged roads in city," Dawn. [Online]. Available: https://www.dawn.com/news/1859688

[3] "Govt celebrates Citizen Portal success despite only 30% satisfaction rate," ProPakistani, Aug. 6, 2026. [Online]. Available: https://propakistani.pk/2026/08/06/govt-celebrates-citizen-portal-success-despite-only-30-satisfaction-rate/

[4] mySociety, "FixMyStreet in the UK." [Online]. Available: https://www.mysociety.org/community/fixmystreet-in-the-uk/

[5] D. Arya, H. Maeda, S. K. Ghosh, D. Toshniwal, and Y. Sekimoto, "RDD2022: A multi-national image dataset for automatic road damage detection," Geoscience Data Journal, 2024. [Online]. Available: https://arxiv.org/abs/2209.08538

[6] National Coordination Office for Space-Based Positioning, Navigation, and Timing, "GPS Accuracy." [Online]. Available: https://www.gps.gov/gps-accuracy (accessed Sep. 4, 2026).

[7] PostGIS Project, "ST_ClusterDBSCAN," PostGIS Reference. [Online]. Available: https://postgis.net/docs/ST_ClusterDBSCAN.html (accessed Sep. 4, 2026).

[8] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in Proc. IEEE Int. Conf. Computer Vision (ICCV), 2011, pp. 2564-2571, doi: 10.1109/ICCV.2011.6126544.

[9] Ultralytics, "Ultralytics YOLO Docs." [Online]. Available: https://docs.ultralytics.com/ (accessed Sep. 4, 2026).

[10] Open-Meteo, "Weather API Features." [Online]. Available: https://open-meteo.com/en/features (accessed Sep. 4, 2026).

[11] Expo, "Location: LocationObject," Expo SDK Documentation. [Online]. Available: https://docs.expo.dev/versions/latest/sdk/location/ (accessed Sep. 4, 2026).

[12] A. Kumar, Sarichand, and R. Bai, "CivicPulse: A civic platform that proves a road hazard was actually repaired," Final Year Project Proposal 23F-90, Dept. of Computer Science, Sukkur IBA University, 2026.

# 7. Appendices

## Appendix A: Traceability

**Table 12: Traceability of objectives to requirements and use cases**

| Objective | Requirements | Use cases |
|---|---|---|
| O1 | FR-11 to FR-14 | UC-05, UC-06, UC-12 |
| O2 | FR-15, FR-16 | UC-05, UC-11 |
| O3 | FR-21 to FR-24 | UC-10, UC-13 |
| O4 | FR-19, FR-20 | UC-11 |
| O5 | FR-17 to FR-19 | UC-15 |
| O6 | FR-25 to FR-29 | UC-08, UC-09, UC-16, UC-17, UC-18 |
| Common base | FR-01 to FR-10, FR-30, FR-31 | UC-01 to UC-04, UC-07, UC-14, UC-19 |

## Appendix B: Initial Parameter Values

These values are starting points. They remain subject to supervisor confirmation and tuning from measured data.

**Table 13: Initial parameter values**

| Item | Initial value |
|---|---|
| Pair threshold | min(ceiling, base + k x mean accuracy), with base 5 m, k = 1.0 and ceiling 30 m |
| Active window | 60 days |
| Severity | Based on the share of the image covered by detection boxes: below 1% gives 1, below 3% gives 2, below 7% gives 3, below 15% gives 4, and otherwise 5. Three or more defects add 1, to a maximum of 5. |
| Priority score | 100 x (0.40 S + 0.25 C + 0.20 T + 0.15 W), where S = severity / 5, C = min(1, ln(1 + n) / ln(11)) for n supporters, T = min(1, days open / 30), and W = zone risk x category sensitivity |
| Weather risk | Forecast rainfall for the next six hours / 25 mm, capped at 1. A warning is issued at a risk of 0.5 or above with a probability of at least 60%. |
| Points | New report 10, confirmation 5, volunteer task 20, repair photograph 30; per-pin cap 100 |